Automatic performance evaluation of the intracorporeal suture exercise.

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Abstract

This work uses deep learning algorithms to provide automated feedback on the suture with intracorporeal knot exercise in the fundamentals of laparoscopic surgery simulator. Different metrics were designed to provide informative feedback to the user on how to complete the task more efficiently. The automation of the feedback will allow students to practice at any time without the supervision of experts. Five residents and five senior surgeons participated in the study. Object detection, image classification, and semantic segmentation deep learning algorithms were used to collect statistics on the practitioner's performance. Three task-specific metrics were defined. The metrics refer to the way the practitioner holds the needle before the insertion to the Penrose drain, and the amount of movement of the Penrose drain during the needle's insertion. Good agreement between the human labeling and the different algorithms' performance and metric values was achieved. The difference between the scores of the senior surgeons and the surgical residents was statistically significant for one of the metrics. We developed a system that provides performance metrics of the intracorporeal suture exercise. These metrics can help surgical residents practice independently and receive informative feedback on how they entered the needle into the Penrose.

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